iDoRemind: A Location-Based Reminder Application for Android

Srihari Reddy Pamulapati
Department of Computing Sciences
Texas A&M University-Corpus Christi
Corpus Christi, TX, USA
sriharireddy09@gmail.com

Longzhuang Li
Department of Computing Sciences
Texas A&M University-Corpus Christi
Corpus Christi, TX, USA
Longzhuang.Li@tamucc.edu

***Abstract*— In the paper, a location-based reminder application *iDoRemind* is developed to assist people in remembering and managing their everyday activities. *iDoRemind* overcomes the weakness of the existing location-based reminder applications by repeated alerting the user and suggesting the alternative locations based on users' preferences and their current location. The suggested locations are ranked according to the calculated *preference-ratio*. In addition, the reminders can be shared in a group or social media. The group option makes it easier for people to manage group activities. It has been shown in the evaluation that users are highly satisfied with the services provided by *iDoRemind*.**

Keywords—location-based reminder, preferences, suggestions, Google Maps

I. Introduction

In the present world, people have to schedule a number of tasks to be accomplished in advance. These tasks vary from the meetings at the work place to the grocery shopping. Since it is hard to remember all the activities, the utilization of reminder systems on smart phones came into place. The time-based reminders are common in the traditional methods, which help to alert users on the particular date and time. The time-based remind applications may not function well if the user is not close to the location at the time specified by the reminder. On the other hand, the location-based reminders without time restrictions may not be useful either if the user travels to that particular location at the wrong time. As a result, the reminders combining both time and location information can serve the user's needs better. This type of reminders triggers the alert when the user reaches a particular location or the specific date and time has come [2].

There are many location based reminder applications available in the market and only few applications make use of time along with the location to remind users [2,6,7,8,9]. The common problem of the existing location-based applications is that if the user misses an alert, there are no alternative solutions offered to help her. It is very common for the user not to travel to a particular location where already planned to visit within a period of time. For example, the user is not be able to travel to or near to the specified pharmacy store to buy medicines, the existing applications fail to provide suggestions for other pharmacy stores close to her current location. There are situations where the user misses to notice the important alerts in her phone. In this case, the reminder should repeatedly remind her about the mattered activities, such as a bill payment or a birthday party. Sharing the reminder with related people and repeated alerting the user are the viable solution in practice. Furthermore, the existing reminder applications do not have an intelligence to come up with a suggestion to set an autonomous reminder even if the user has shown her preferences in a number of her previous reminders.

Some shortcomings of the existing location-based reminder applications can be improved by utilizing the geofencing technique, which creates a virtual boundary around a point using Global Positioning System (GPS). When a device enters this boundary, it will trigger the reminder. This can help in generating more information about the user according to her preferences and the specific reminders [3]. This location-based reminder along with the information about the surrounding places will help to provide more effective reminders [4]. It will give a picture about interests and reminder patterns by continuously evaluating the user interests. This can be done by analyzing the data of the user. It requires an efficient database that can store all the data without affecting the performance of devices [5]. Sharing the reminder with your family members or roommates can be very useful. *iDoRemind* in the paper is a mixture of all these features.

In the paper, a new location-based reminder application, *iDoRemind*, is developed to address the limitations in the existing reminder applications. The useful and unique features offered by *iDoRemind* include preference options, nearby place suggestion, repeat reminding, and sharing the reminder in a group. To be more specific, *iDoRemind* allows users to create location-based reminders, provides features to make the reminders date and time restricted, is able to suggest the locations to users based on current preferences and reminders, assigns each user a unique code so that the user can be added to other groups, automatically alerts the members in the same group if any person in her group travels near the location in the reminder, shares the reminder with other people in the same group when the reminder remains unattended, and alerts the user when she travels near similar kind of stores to that in the reminder.

II. Existing Applications

There are many reminder applications available for android phones but very few of them can work based on the

978-1-5386-2074-8/17 $31.00 © 2017 IEEE
DOI 10.1109/FiCloud.2017.9

IEEE computer society

location. In addition, the existing location-based applications still have many limitations. Some of the location based reminder applications and their shortcomings are discussed as follows.

TABLE I. COMPARISON OF *IDOREMIND* AND THE EXISTING LOCATION-BASED APPLICATIONS

Features \ Applications	Location Reminders	NTRT	Google Keep	Place-Its	Remind me	iDoRemind
Reminder based on location	✔	✔	✔	✔	✔	✔
Time restriction	✔	✕	✕	✕	✔	✔
Date restriction	✕	✕	✕	✕	✕	✔
Repeat reminder	✕	✕	✔	✕	✕	✔
Groups	✕	✕	✕	✕	✕	✔
Preferences and suggestions	✕	✕	✕	✕	✕	✔
Nearby places	✕	✕	✕	✕	✕	✔
Sharing via social media	✕	✕	✔	✕	✔	✔
Delete reminder	✕	✔	✔	✔	✔	✔
Picking location on map	✔	✕	✔	✔	✔	✔

Location Reminders [8] is an android application that reminds users about reminders based on location. The application can create reminders whether to alert at the time of entry/exit with the time-based restriction in the morning/afternoon. The application cannot customize the starting and ending days as well as the hours, does not support the reminder sharing, and does not have the option to enter the radius value. Furthermore, the location cannot be deleted from list once it is added.

NTRT (Next Time Remember To) location reminder [9] is another android application which works based on location. *NTRT* offers live status of distance to the location and is easy to search and select locations. But *NTRT* provides no time restriction, has no option to enter a reminder at the current location, and cannot share a reminder and select locations on map.

Google Keep (GK) [7] is a note taking and reminder application by Google. The reminders are based on either time or location but not both. GK synchronizes reminders with all the devices registered under the user's Google account and can share reminders with others. The problems with GK are that it has no time restriction for location-based reminders and has no option to customize radius. In addition, unorganized user interface makes it difficult to use.

Place-Its [6] permits the integration of location-based reminders into people's daily practice. The advantages of this application are that the user can pick the location on the map and create the reminder at the chosen location. The application triggers the reminder when the user enters/exits the area. Place-Its has relatively low location accuracy, and lacks good features offered by *iDoRemind* in this paper.

RemindMe [2] is a location based application with various location tagging options using social networking APIs. Although *RemindMe* supports location detection and tagging via Google Maps and Foursquare, and adds time restriction to the location, it does not support functions, such as preferences and nearby locations suggestion, offered by *iDoRemind* in the paper.

The comparison between *iDoRemind* and the existing location-based applications is shown in Table 1. *iDoRemind* supports all the listed features, while the existing applications only support part of them.

III. *IDOREMIND* SYSETM DESIGN

The architecture of *iDoRemind* is shown in Figure 1. *iDoRemind* consists of presentation layer, business logic, rest access layer, shared preferences, location manager, Google Places API, Google Maps API, and the Web server. The Web server is utilized to establish the connection with rest access layer as well as with the centralized database.

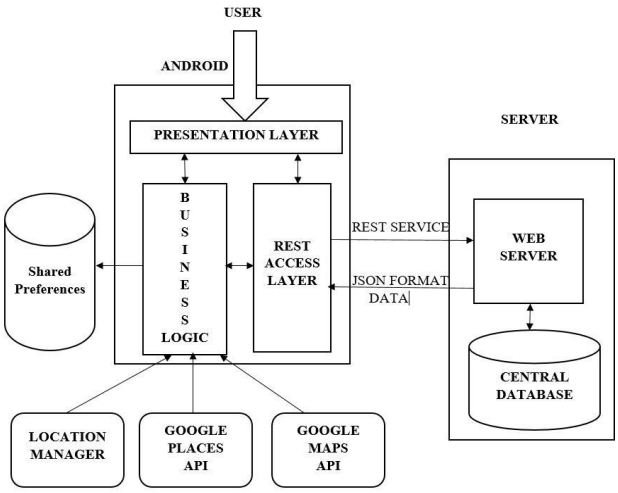


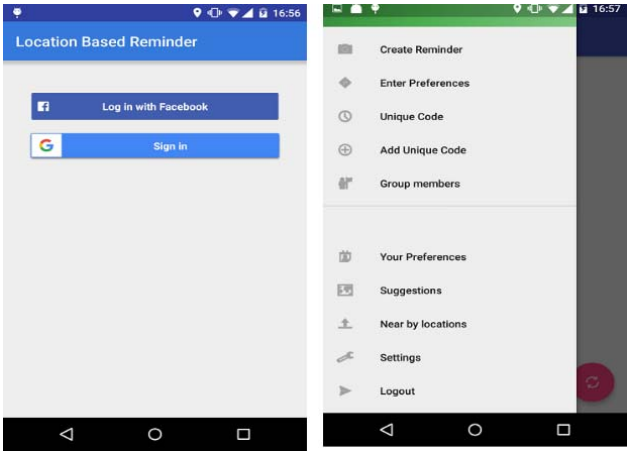
Fig. 1. The architecture of *iDoRemind*.

In Figure 1 the presentation layer contains the user interface, and the rest access layer is used to establish communication between the Web server and users. When a

user makes a request to create a new reminder, the request is processed by the Web server on Amazon Web Server (AWS), and then a row is inserted to the reminder table in the centralized database. The reminder table contains the detailed information about the new reminder, including the reminder title, location information (latitude, longitude values), time restrictions, and other users (or groups) that the reminder is shared with. Locations and location related information are obtained via the location manager, Google Places API and Google Maps API. Location manager is employed to obtain the user's current location, Google Places API is used to get nearby places, average rating and cost range, and Google Maps API allows the embedding of Google Maps onto *iDoRemind* using a simple JavaScript interface or a Flash interface. The shared preferences component stores users' detailed information, such as user ID and name as well as their personal preferences. The business logic creates the request to AWS based on the details obtained from the shared preferences, and processes the data from Google Places and Google Maps APIs.

A. Create Reminders

When the user launches *iDoRemind* for the first time she needs to sign in and authorize the usage of mobile phone resources. The user can login into *iDoRemind* through her Facebook or Google account so that she does not have to create and remember a new user account and password. Figure 2(a) shows the login screen and Figure 2(b) the dashboard of *iDoRemind*.



(a) (b)

Fig. 2. (a) Login screen and (b) dashboard of *iDoRemind*.

After logging into *iDoRemind*, the user can create a new reminder by entering the title and picking a location from Google Maps or typing the text address of the location. The user has the options to restrict the reminder based on either time, date or both. If the user needs a particular reminder to be launched periodically, she can set the period using the *repeat* option. For example, if the user needs to collect the paycheck every two weeks, she can set the reminder to repeat for every fourteen days. Figure 3 shows an example on creating a new reminder by picking a location from Google Maps, where the location the Shores Apartments is chosen. Two reminders with/without restrictions are shown in Figure 4. The first reminder in Figure 4 is restricted between 7:05 am to 7:06 pm from 11/29/2016 to 1/02/2017 and is also set to repeat every fourteen days. The second reminder does not contain any restrictions.

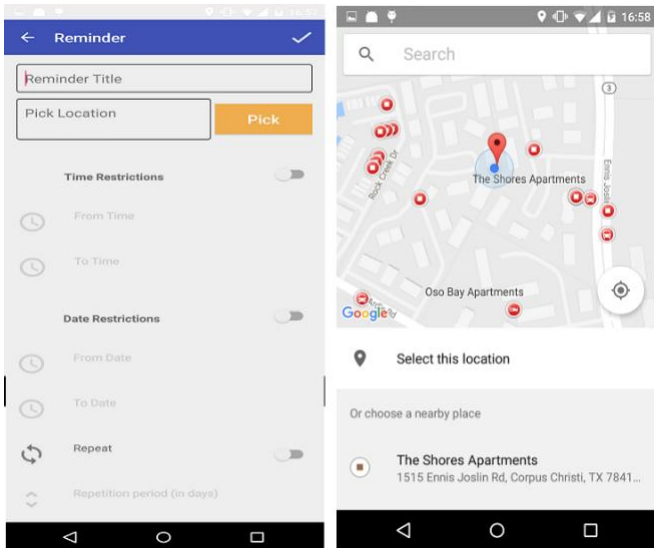


Fig. 3. Create a new reminder.

Once the reminder is created, the user can share the reminder with her family/friends using the *Assign Users* button (see Figure 4) by grouping multiple users into a group. *iDoRemind* alerts every group member if anyone in the group travels to the location in the reminder. She can also share the reminder on the social media (e.g, Facebook and Gmail) by clicking the *Share* button (see Figure 4).

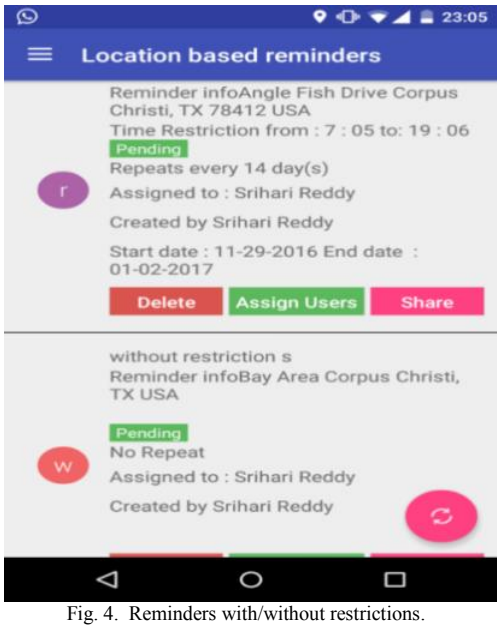


Fig. 4. Reminders with/without restrictions.

B. *Preferences, Suggestion, and Nearby Places*

The user can select her preferences by checking the boxes among the options provided by *iDoRemind*. In the paper the preferences refer to the places that the user may visit frequently. Once selected the user's preferences are updated in the Shared Preferences component. Figure 5 shows an example that the user chooses two favorite locations, café and library, as her preferences.

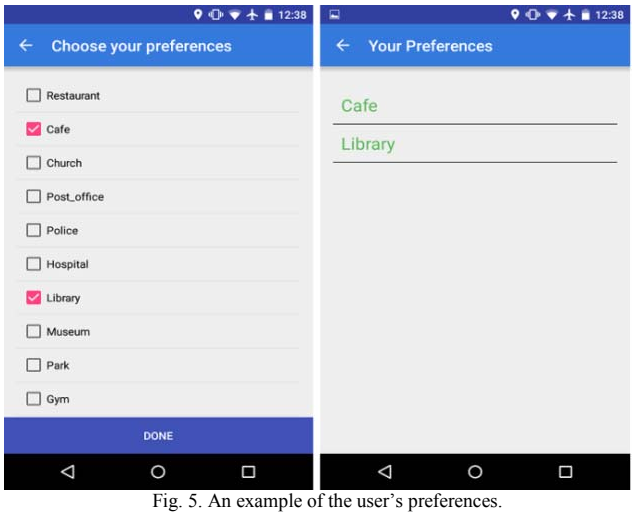
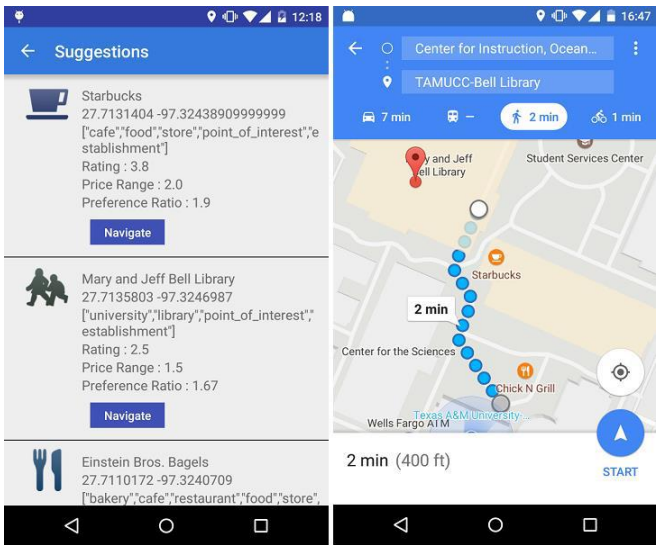


Fig. 5. An example of the user's preferences.

Based on the preferences selected by the user *iDoRemind* autonomously suggests some places that she might want to visit. The suggested places are generated by matching the preferences against the location lists from Google Places API according to the latitude and longitude values from location manager. For example, the user can check suggestions for the cafés around her (e.g., with a radius of 400 meters) when she travels to some new place if she selects café as her preference. The suggested locations are displayed in the descending order of the *preference_ratio* values, which are calcuated as:

$$preference_ratio = average_rating \,/\, cost_range \qquad (1)$$

where both *average_rating* and *cost_range* are retrieved from Google Places API. *average_rating* contains the place's rating with a value range from 0.0 to 5.0 based on the aggregated user reviews. *cost_range* is obtained by converting the counting of the dollar signs into an integer. To be more specific, $ is considered as 1, $$ as 2, and $$$ as 3. The default values for *average_ rating* and *cost_range* are 2.5 and 1.5, respectively. The user can navigate to the location by clicking on *navigate* button in the suggestions. Figure 6(a) shows the examples of the suggested locations according to the user's preferences, such as café and library. Figure 6(b) presents the navigation from the current location *Center for Instruction* to the suggested *location Mary and Jeff Bell Library* at Texas A&M University-Corpus Christi once the user decides to go to the library.



(a) (b)

Fig. 6. Suggestions based on the preferences café and library.

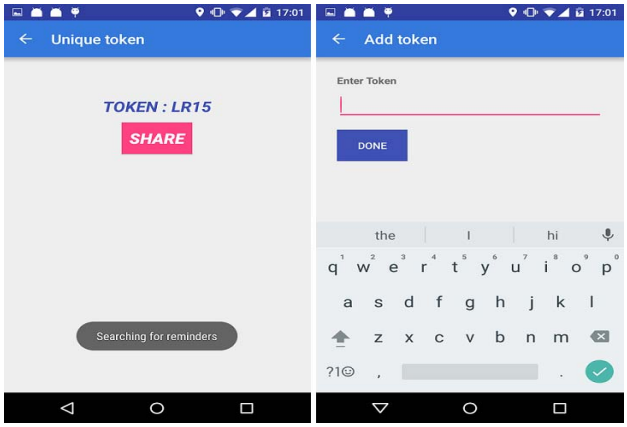
The *Nearby Places* option provides location suggestions based on the some reminder instead of preferences. For example, if the user wants to go to some grocery store but for some reason she is not able to travel to that location, she can check the grocery stores nearby in this option. This option helps in finding the alternatives and thereby saves time.

Some users might not like the notifications about suggestions from *iDoRemind*. In such case, they can simply turn off the preference-based suggestions and nearby places notifications in the settings.

C. *Unique Token and Groups*

In *iDoRemind* each user is assigned an unique token, which represents the user's exclusive code that she can share with her family or friends when to form groups. The groups work in one-way fashion such that user *y* is in user *x*'s group if user *x* adds the token of user *y*. Similarly, user *x* cannot be in user *y*'s group unless user *y* adds the token of user *x*. Figure 7(a) shows an example of the unique token (e.g., LR15) that the user can share with other users and Figure 7(b) is the screen for the user to add other users to her group. After the group is formed, the user can share her reminders with the picked members in her group. If any person in the group travels close to the location in the reminder, all the members in the group are alerted automatically.

The reminders can also be shared in different external applications, such as WhatsApp, text messaging, messenger, and Gmail, through the unique token. Sharing of a reminder in WhatsApp is shown in Figure 8, which is also realized by sharing the user's unique token via the application.



(a) (b)

Fig. 7. (a) Unique token and (b) add token screen.

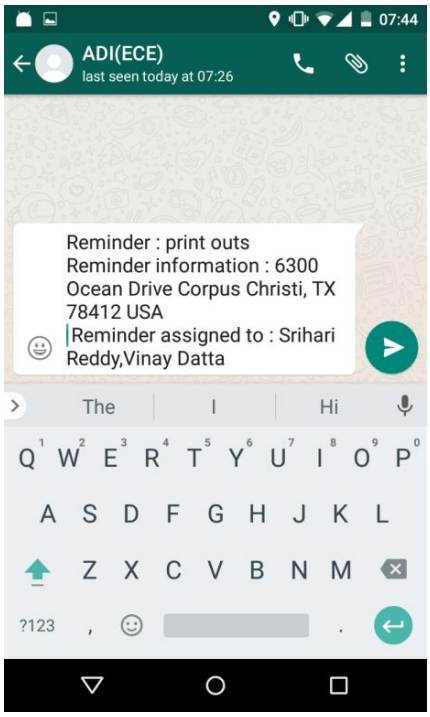


Fig. 8. Sharing of a reminder in WhatsApp.

IV. Evaluation

The developed system tested using Nexus 4 mobile phone with android version 5.1.1. This application is compatible for android 4.0 and later version.

In order to evaluate *iDoRemind*, the evaluation is collected from fifteen users who used the application from one week to fifteen days. The evaluation consists of thirteen questions developed from Technology Acceptance Model [1,2]. Questions 1-7 measure the perceived usefulness, questions 8-11 the perceived ease of use, and questions 12-13 the user satisfaction. The evaluators have the option to rate each question between 0 and 10, where 0 indicates strongly disagree and 10 indicates strongly agree. In the evaluation, the users are encouraged to use all the features offered by *iDoRemind*. Table 2 shows the list of questions in the evaluation and the average rating for each question.

As it can be seen from Table 2, Sharing the reminder via social media (Q6) was rated the lowest (8.4) because the following two major reasons, (1) the participants in the study did not want to over expose their own activities on the social media and (2) some participants rarely used any social media. Suggestions based on preferences (Q4) received the second lowest rating (8.6) since we did not offer a large variety of preferences for the users to choose in the current implementation. All other questions in the evaluation achieved above 9.0 scores. Especially the *group* option (Q3) was given the highest rating (9.8). The participants found the group reminder option very interesting and useful because no other reminder applications provide the similar function. It can be seen that *iDoRemind* is very easy to learn and use from the ratings of questions 8-11. The responses of questions 12-13 clearly demonstrated that *iDoRemind* can meet users' needs and provide useful functions.

TABLE II. List of Questions in the Evaluation and Their Average Ratingns

	Question	Average rating
1.	*iDoRemind* gives notifications as expected.	9.73
2.	Applying restriction to a reminder is useful.	9.07
3.	*Group* option is useful.	9.8
4.	Suggestions based on preferences are useful.	8.6
5.	*Nearby places* option is useful.	9.07
6.	Sharing the reminder via social media is useful.	8.4
7.	Sharing the reminder with others when the user cannot attain (clicking on *Ignore* button) is useful.	9.2
8.	The user interface is easy to understand.	9.5
9.	Reminding alerts are clear.	9.27
10.	Navigating the application is easy.	9.5
11.	Setting up the tasks is easy.	9.43
12.	*iDoRemind* application is satisfying.	9.5
13.	I would recommend *iDoRemind* to other people.	9.73

V. Conclusion and Future Work

In the paper, a location-based reminder application *iDoRemind* is developed for smart phones and tablets with the Android operating system. In *iDoRemind* the reminders can be restricted with date and time and be set to repeat for the particular period of time. The unique feature of *iDoRemind* is that the user can create her own group and share her reminders to the selected members in the group or with the whole group. The reminder is triggered if any assigned member in the group travels near the location in the reminder. In addition, locations are suggested based on the user's preferences and the nature of

the location in the reminder. In the evaluation *iDoRemind* achieved highly favorable feedbacks.

iDoRemind can be improved by adding more functionalities. First, the reminders should be more intelligently monitored to reduce the battery consumption. For example, the reminders with locations far away can be kept idle. Second, the personalized preferences can be automatically inferred by using the data mining techniques on the frequently visited locations. Third, *iDoRemind* can recommend the potential group members based on users' activities and preferences.

ACKNOWLEDGMENT

This work was supported in part by grants from NSF MRI 1429518.

REFERENCES

[1] F. D. Davis, "Perceived usefulness, perceived ease of use,and user acceptance of information technology," *MIS Quarterly*, 13(3): 319-340, 1989.

[2] A. M. Ertugrul and I. Onal, "RemindMe: an enhanced mobile location-based reminder application, in *Proceedings of the International Conference on Future Internet of Things and Cloud*, 2014.

[3] S. R. Garzon, B. Deva, G. Pilz, and S. Medack, "Infrastructure-assisted geofencing: proactive location-based services with thin mobile clients and smart servers," in *the IEEE International Conference on Mobile Cloud Computing, Services, and Engineering*, 2015.

[4] Y. Li, A. Guo, S. Liu, Y. Gao, and Y. Zheng, "A location based reminder system for advertisement," in *Proceedings of the International Conference on Multimedia*, 2010.

[5] K. S. Rao and M. K. Murthy, "Mining user interests from web log data using long-period extracting algorithm," *International Journal of Applied Engineering Research*, 10(10):25017-25026, 2014.

[6] T. Sohn, K. A. Li, G. Lee, I. Smith, J. Scott, and W. G. Griswold, "Place-Its: a study of location-based reminders on mobile phones," M. Beigl et al. (Eds.): *UbiComp* 2005.

[7] Google Keep, Retrieved from https://play.google.com/store/apps/details?id=com.google.android.keep&hl=en. Retrieved in December, 2016.

[8] Location Reminders, https://play.google.com/store/apps/details?id=org2.nick_studio.placebox&hl=en. Retrieved in December, 2016.

[9] Next Time Remember To, https://play.google.com/store/apps/details?id=com.enselta.ntrt.android&hl=en. Retrieved in December, 2016.